



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Candy Match Saga: Candy AI Lab

"Where Every Match Teaches AI Something New."

CSA0920 – Programming in Java

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GAME CONCEPT OVERVIEW

What is the Game?

- AI-inspired Match-3 puzzle game.
- Player acts as a Candy Scientist.
- Match candies to discover new candy evolution rules.
- Train an AI Assistant by teaching new candy rules.
- Complete laboratory missions to progress.

Win Condition

- Complete all laboratory missions.
- Unlock advanced candies.

Lose Condition

No valid moves or all moves are exhausted.

REFERENCE GAME STUDY

Reference Game: Candy Crush Saga

Mechanics Reproduced:

- Match-3 gameplay.
- 8×8 candy grid.
- Score calculation.
- Goal-based progression.

Simplified Features:

- Single-level gameplay.
- Java Swing desktop version.
- Local MySQL database.

New Features Added:

- AI Learning Assistant.
- Custom Candy Creator.
- Candy Research Book.
- Goal Tracker

OBJECTIVES /LEARNING OUTCOMES

- Implement 2D Grid Manipulation.
- Apply OOP concepts.
- Use Java Collections.
- Develop Java Swing GUI.
- Implement AI Rule Learning.
- Save progress using JDBC.
- Handle exceptions.
- Build an interactive puzzle game.



PROPOSED MODULE BREAKDOWN



Module 1 – Candy Evolution Engine (OOP)

Handles candy evolution rules.

Creates new candies after successful matches.

Module 2 – Smart Candy Grid Manager (Collections)

Manages the 8×8 game board.

Handles candy placement, movement, and matching.

Module 3 – Interactive Game Interface (Java Swing)

Displays the game board.

Handles click, drag, and drop interactions.

Module 4 – AI Knowledge & Mission Manager(HashMap & Queue)

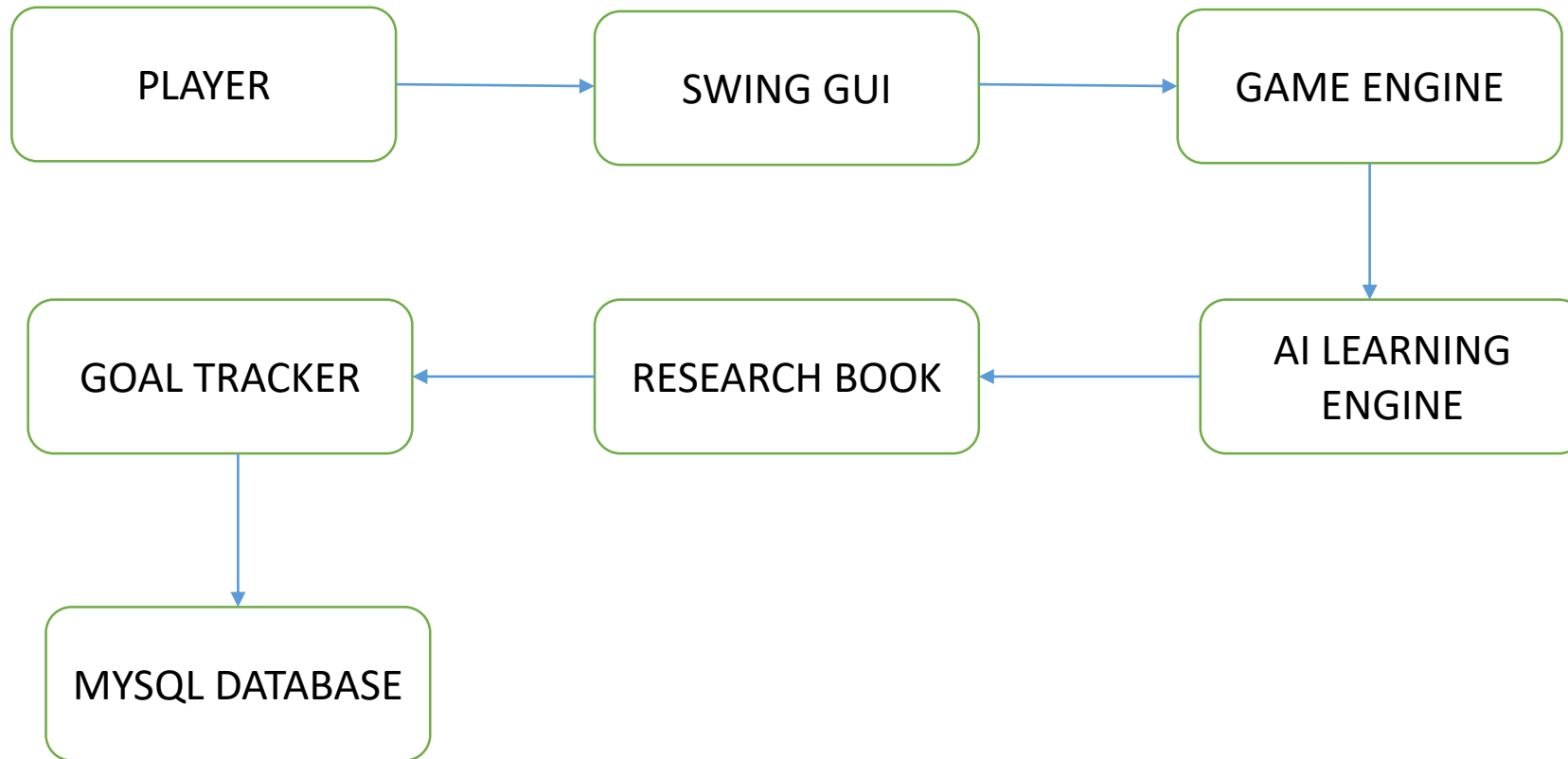
AI learns new candy rules.

Tracks player missions and provides hints.

Module 5 – Player Progress & Database Manager (JDBC)

Saves player score, AI level, discovered candies, and missions

SYSTEM ARCHITECTURE



TECH STACK & WHY

Technology	Purpose
Java	Game Logic
Swing	GUI
OOP	Candy Objects
Collections	Board & AI Rules
JDBC	Database Connectivity
MySQL	Store Player Progress
Multithreading	Timer & Animations
Exception Handling	Error Management

PROPOSED DATABASE SCHEMA

Database: candy_ai_lab

Table: player_progress

Column	Type
Player_id	INT
player_name	VARCHAR(50)
score	INT
moves	INT
Ai_level	INT
learned_rules	TEXT
Discovered_candies	TEXT
Completed_goals	TEXT
Save_date	DATETIME

MULTITHREADING PLAN

Threads Used:

- Game Timer Thread
- Candy Animation Thread
- AI Hint Thread
- Background Database Save Thread

Benefits:

- Smooth animations.
- Faster AI response.
- No GUI freezing.
- Background save operation.

EXCEPTION HANDLING PLAN

Possible Exceptions

- Invalid candy merge.
- Duplicate candy rule.
- Database connection failure.
- No valid moves available.

Solution

- Use try-catch blocks.
- Display user-friendly messages.
- Prevent invalid operations.

WORK PLAN/TIMELINE

Weak	Task
Weak 1	GUI Design & Board Creation
Weak 2	Matching Logic & AI Learning
Weak 3	JDBC & Research Book
Weak 4	Testing & Final Presentation

ANTICIPATED CHALLENGES

Challenges:

- Managing AI learning rules.
- Efficient 8×8 board handling.
- Preventing invalid candy merges.
- Synchronizing animations.
- Database consistency.

Proposed Solution:

- Java Collections.
- Exception Handling.
- Multithreading.
- JDBC Transactions.



Engineer to Excel



THANK YOU